

Perfectly Imperfect: A Speculation about Wabi-Sabi Inspired User Experience Design

Wabi-Sabi Inspired UX Design

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ABSTRACT

Businesses and the software built for them are usually associated with models that have economic benefit at their core. The various disciplines of design, such as communication design for branding or design for human computer interaction (HCI), support those value-creating processes. Wabi-Sabi, a Japanese philosophy and aesthetic concept that is not rooted in creating monetary value, approaching a discipline within HCI, or vice versa, therefore asks for pathways that allow open, interdisciplinary and intercultural learning. We used speculative and narrative methods such as poetic writing and scriptwriting to prototype and create a narrative to illustrate our concepts. This paper investigates how Wabi-Sabi principles could be applied to the user experience of digital products, specifically business software. It further highlights a methodological approach that has proved valuable in investigating two initially contrary themes.

CCS CONCEPTS

• Human computer interaction (HCI); • Enterprise applications; • Interaction design theory, concepts and paradigms;

KEYWORDS

Wabi-Sabi, User Experience Design, Hypothesis based structured brainstorming, Transience

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1 INTRODUCTION

Wabi-Sabi is a traditional Japanese aesthetic concept that describes the “beauty of things imperfect, impermanent, and incomplete” [6]. In recent years, it has been increasingly discussed and applied to

fields like architecture and interior design, both in Japan and abroad [9].

The application of Wabi-Sabi concepts to software products has not yet been extensively discussed. Business Software is a product category that supports companies of all sizes in the running of their business processes, such as finance, control, supply chain (in production industries) and patient management (in medicine). Because of the way businesses operate today, business software emphasizes productivity and goal or target achievement, and helps users focus on the next task that needs to be taken care of. The Wabi-Sabi philosophy has the potential to put the human mind in an introspective state, which could have a calming or inspiring effect. We want to investigate alternative ways to design thriving enterprise product experiences and work cultures and think that Wabi-Sabi could help enable this.

Because Wabi-Sabi is a concept that cannot be defined in simple, concrete terms, we have adopted an abstract style of writing in our research, to retain the undefinable quality of this aesthetic and communicate its essence without overly defining it.

1.1 Three variations on the experiences of Wabi-Sabi

Space. Lots of white space, a porcelain background, fluent and adaptive, to give contrast to the words and numbers that build up. Signals, almost audible in their visual intensity, pop out in strong colours, demanding attention, drawing focus, and instigating a drilling down and following up on something overdue, over-budget, or understaffed. It might be wise to act upon those signals. To chase overdue purchase orders for stocking our pharmacy, to clarify why there has been more off-contract than budgeted spending in the last quarter, or to reschedule next week's appointments because half the staff is on leave. It feels clear, structured and efficient; even beautiful. “Do something” it says. So, follow the call, tick off the items on the list, be productive, get the job done. Once one item is ticked off the next awaits, leaving little memory of what was done and how. Does it matter? A feeling of momentary relief, for sure, but could there be more?

Space. The photographer lets us explore the image, in which is choreographed a hierarchy of layers for us to view: foreground, background. It is up to the viewer to explore, to adapt, to find their own focus. It is not the darker, upper two-thirds of the image that dominate, but the lighter, lower third – unusual for the western viewer, but soothing, calming. As the house is approached, the stones in front show “simple, natural materiality” [14].

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Figure 1: Walter Gropius photographing the tea pavilion at Villa Katsura, Kyoto, Japan. Bauhaus-Archiv Berlin [13]

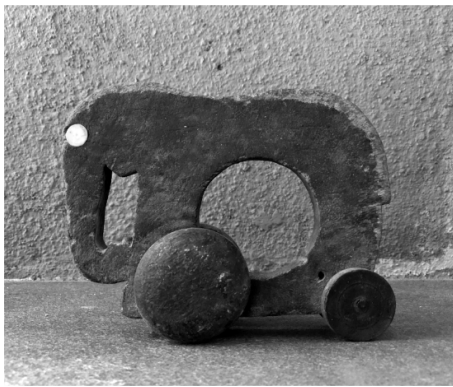


Figure 2: Toy elephant made of wood. Date and crafts-person unknown, purchased from the Darbargadh area, Gujarat, India. Photo: Aparna Kongot

The man photographing the tea pavilion of the Japanese Villa blends into his surroundings, yet guides the viewer to the exterior structures and facade of the pavilion. Building blocks, symmetrical, but varying in their materials and patterns, and their shades of light and dark, lead our gaze through to the other side of the pavilion. The thatched roof, in contrast to the smooth, lower structures of the building, mediates the handover and transition from human craft to nature, in the trees surrounding the pavilion. Thirty-five years earlier in a German city, the person in the image founded a school of art and design that influenced subsequent schools and designers with more than just its key design principle of simplicity. Here, in 1954, he seems to reinstitute the exchange of knowledge and inspiration from different regions of the world, through their cultures and their aesthetics. Appreciating, honouring, adapting and honing what was seen, heard, conveyed, shared and experienced, including the concept of “minimalism” [12].

Space. The need for space to play. A glance over the rim of the computer display reveals childhood memories. A shape resembling an elephant, whose form contains no straight lines. Rather, all is one flow, rounded, soft, no hard edges, yet irregular, scratched and weathered, showing how it has aged. Two different sets of wheels, the front axle larger than the one at the rear, giving it

an initially imperfect appearance and leading the view to wonder whether the unknown, and thus anonymous, [6] craftspeople had a reason for creating it this way, or whether the user has adapted it through interaction. It transmits a “charm of playful humor” [8], an unspoken invitation to put our fingers through the circular opening and start pushing it to play. The circle that resembles a pie chart, depicting the product categories we ordered, flawless, clear, forcing our focus to return to work from where we drifted off with our thoughts.

1.2 Speculations based on an intuition

Missing as it does a clear problem statement, such as a user’s challenge or a specific need, and following no more than intuition, what added value could Wabi-Sabi have for user experience (UX) design and UX user research? Is there a business value in investing in Wabi-Sabi concepts? Critiques allege that Wabi-Sabi is (just) a “Japanese philosophy” [12], describing the beauty of things as “imperfect, impermanent, and incomplete” [4] and question its successful and viable application. The goal of our investigation thus became to understand to what extent its concepts could enhance the experience of today’s digital products for its users.

Peter Boermsa [3] was among the first authors to investigate how Wabi-Sabi concepts could be leveraged for web design, by utilizing “personalisation, [the] separation of content [or] small pieces, loosely joined”. While purposefully creating a Wabi-Sabi based internet site is impossible, Sherwin [11] states, the infusion of certain Wabi-Sabi elements could spark motivation for design practitioners to get interested in the philosophy. Tsaknaki and Fernaeus note that although key aspects of Wabi-Sabi in Human Computer Interaction (HCI), such as the impermanence or crudeness of an object, may initially appear negative, they can also “satisfy a desire for more authentic interactive products” [12]. The authors suggest the consideration of three design principles: sustainable design, to foster the interactive use of products over a long period of time; allowing designs to be unfinished; and using technological shortcomings as a starting point for new approaches and ways of thinking. The notion of sustainability as an important design factor is similarly highlighted by Lobos [7]. Salvia et al. [10] confirm that failings or deficiencies can spark innovation for the maker, and invite the user to develop a stronger emotional tie to the product, due to the uniqueness they add.

While previous research engaged Wabi-Sabi concepts in the use of and alignment with digital and interactive products, ours aims to add new findings from the perspective of standardized business software.

1.3 Building hypotheses

The literature review unveiled a sufficient number of Wabi-Sabi principles [6, 12] that we could enrich to become ‘Wabi-Sabi components’, such as ‘celebrating imperfection’, ‘care needs to pass for quality to get aware’ or ‘mutual creation between humans and nature’. In the first step of our research, these components were each assigned initial hypotheses to act as an allocable and speculative pair for an initial ideation about how the component could be realised in a business systems user interface. This hypothesis-based structured brainstorming enabled us to extend possible outcomes of

our initial hypotheses [1], using the constraint of a strictly-focussed compatibility with Wabi-Sabi principles.

The next step aimed to generate more precise clusters of all results of the ideation process which led to 29 clusters assigned to 7 Wabi-Sabi components. During the process of clustering it became apparent that the initial hypotheses needed to be reframed to be more precise and also to overcome the limited variance of initial hypotheses due to the small number ($n = 2$) of researchers involved in the process [1]. The proficiency of Wabi-Sabi components, hypotheses and clusters, along with its ideas, enabled speculation into various contexts and scenarios of how the philosophy could be utilised for business software user interfaces.

2 SCRIPT WRITING AS A FORM OF CONCEPT EXPRESSION

Storytelling is a common tool in the field of Human Computer Interaction for establishing an understanding of the gaps or problems in existing concepts and communicating novel ideas to solve these problems [2]. This technique is effective in sharing a vision to an audience, at a time when the concepts are still too early to be visualized as digital user interfaces. Script writing is one form of storytelling that we have adopted to convey our ideas.

In the following, a fictitious script will be used to speculate future ways of integrating Wabi-Sabi concepts into the design of user interfaces for business software. Script writing is a tested technique used to communicate an idea in the form of a visual story. Visual stories have the power to convey the essence of an idea, while also allowing exploration beyond the idea towards the interpersonal exchanges between characters, the nuances of their relationships, and the emotions that come into play.

Inspired by the technique and its potential, below are scenes that are written to express the influence of Wabi-Sabi on digital user experiences. To effectively communicate the protagonist's characteristics, the stories need to be shaped "in a way that both expresses [the designer's] vision and satisfies the audience's desires." [5]. The scenes we created point to the relationship between humans and technology, the responsibility of the digital product providers and, more importantly, the derived Wabi-Sabi UX concepts. We propose these concepts as helpful insights UX design practitioners can use to design products that embody the experiential qualities of Wabi-Sabi.

Below we have identified five Wabi-Sabi experiences that can be mapped to the UX design of digital products. We propose UX concepts that are derived from each of these Wabi-Sabi experiences and have included abstract illustrations to support this narrative.

2.1 Wabi-Sabi amplifies emotional connection

The Scene: At the Paediatrician's office, Medkit surprises Sam.

On the faded green wallpaper are printed smiling animals. A clock on the wall reads 9 AM. Sam, a 40-year-old paediatrician, greets his assistant cheerfully from his desk. He opens 'Medkit', a medical app on his tablet, to kickstart his day.

Medkit: Congrats, you are a Maker!

Sam: Huh?

Medkit: You informed us via the app that you spend half an hour each morning studying the medical history of that day's patients.

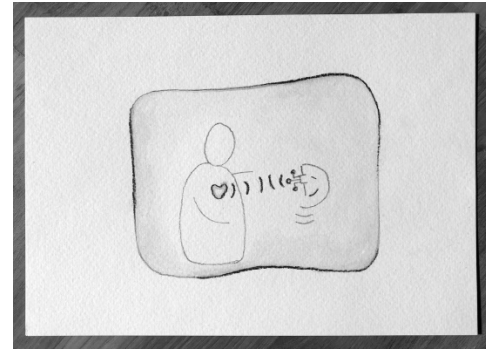


Figure 3: Wabi-Sabi amplifies emotional connections

Your work ethic is our inspiration, and we have turned your habit into a product feature called 'Daily-brief'. Here it is.

Sam (impressed): Wow, that is one useful screen!

Medkit: Here is a 'Maker' badge! Add it to your profile to showcase your contribution to our network of over 5000 doctors. You have also earned a 50% discount on your upcoming annual subscription!

Sam (happy): Heh! Now, who would have seen that coming?

Medkit: A team of 20 product experts, engineers, designers, content writers and quality testers from Germany and India have built this feature over the past 6 months. They send you their best. Enjoy your 'Daily-brief'!

Sam (happy): Sweet!

Medkit: Did you know you have reached a milestone? The next patient will be the 100,000th patient you treated with us.

Sam: Unbelievable, how time flies! Got to start work now. 'Daily-brief' shows a packed day ahead!

Assistant: Yeah! It's a good thing the candy box is already stocked up for our little patients.

2.1.1 Derived Wabi-Sabi UX Concepts. Wabi-Sabi can create positive value and memory reinforcements for the user, by:

- Learning from user habits
- Celebrating the user as a maker
- Surfacing crafting efforts
- Planting positive memory triggers
- Designing an experience, not just a product

2.2 Wabi-Sabi celebrates imperfections

The Scene: Near the examination table, Medkit tests Sam's patience.

An anxious 8-year-old girl and her mother leave Sam's office. A bright LED lamp shines over an empty examination table. Sam focusses on his tablet, looking at an image of his patient's arm with a rare skin condition.

Sam: Add this picture to the 'Quick Bits canvas' of my patient.

Medkit: Opening 'Quick bits' – a fluid and flexible canvas. Added the picture next to an older picture from her previous visit.

Sam: (Draws a circle on the image, writes a note, adds audio recording of his thoughts and links it to the picture, then arranges the layout to suit his purpose). I remember another patient with

a similar skin condition. Can you link the ‘Quick bit’ canvases of these two patients? I need it for my research.

Medkit: Sure. This will take a few seconds . . .

Sam: Okay.

Sam (frustrated, waiting): Well, this is taking longer than expected!

Medkit: We have failed to meet your expectations. Please allow us to explain. It will only take 30 seconds.

Sam (frustrated but curious): Alright.

Medkit: Thanks for giving us a chance to explain. As a result of a recent government policy, we are making technical changes to our product to give you the best experience, and clearly we can do a better job. Your patience helps us deal with the situation while looking after the well-being of our diligent employees who are working tirelessly to get things running smoothly for you. Meanwhile, we will re-attempt to complete your task until we succeed. If you would like a detailed explanation, please leave your number and we will call you back.

Sam (calmer): Hmm . . . I’ve got an idea. How about a new ‘Quick-bit canvas’ for research where I can quickly paste the links to both canvases? This will just take me a minute.

Sam (curious): (creates new canvas) Hey, this is interesting! I can actually use the inbuilt ‘share’ functionality of the ‘Quick-bit canvas’ to share this finding with my research partner. I should share my new find with Medkit!

Medkit: We are thrilled to see how our users discover new ways to use our product in various contexts! This helps us and our users grow together.

2.2.1 Derived Wabi-Sabi UX Concepts. Imperfections can contribute positively to the experience of both user and maker, by:

- Allowing user made fluid layouts
- Surfacing imperfections for honest user experiences
- Breaking the taboo of imperfections
- Embracing a failure culture in order to grow
- Learning from user hacks to tackle failure

2.3 Wabi-Sabi embraces modest aesthetics

The Scene: At Lunch, Sam has a sentimental moment.

While people walk around with plates, hushed conversations fill the hospital’s cafeteria. Sam is alone at his table, eating a salad and casually browsing Medkit.

Sam: I really like the colours on this app.

Medkit: Muted, low contrast, earthy colours, matte finishes, rough, organic and asymmetric layouts is our visual design theme. Our designers believe that modest aesthetics can have a calming effect, which encourages reflection and sparks curiosity. We don’t seek the limelight, but we want to help our users find theirs.

Sam: Hah, sometimes it does feel like I’m walking in the woods with my thoughts!

Medkit: A quiet moment amidst all the chaos is valuable.

Sam: Don’t take this the wrong way, but the app feels incomplete, and I find that refreshing and empowering. It invites me to experiment with the assurance that I can’t ruin anything if I move a few things around.

Medkit: That’s our goal.

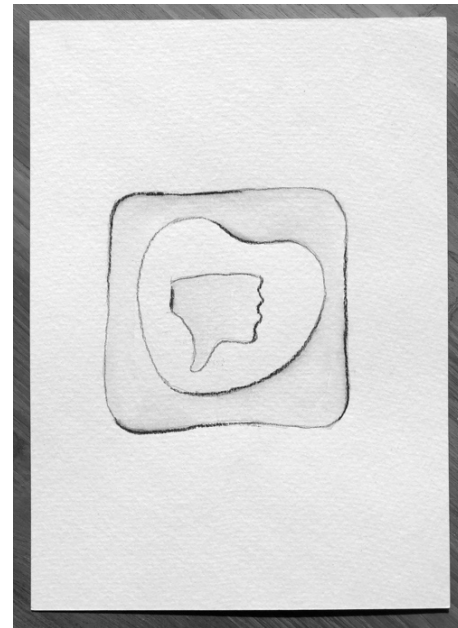


Figure 4: Wabi-Sabi celebrates imperfections

Sam (happy): But my favourite bit, I must say, are the subtle visual organic patterns that look like folds and keep changing in the app’s wallpaper.

Medkit: Each fold is like a wrinkle. It is a testimony to your experience, and as the number of patients you help grows, a new fold is added to the app’s background, or wallpaper, as you call it!

Sam: I thought so! Each fold is a patient I’ve treated. It feels more meaningful than any certificate or award I have ever received.

Medkit: Yes, the folds are generated from patient data. When patients view your profile, the beautiful folds tell them about your years of service, and help build trust.

Sam (amused): I get it. If I were a book, I’d rather have dog ears, yellowish pages and underlined text than remaining crisp with clean white pages!

Medkit: Absolutely.

2.3.1 Derived Wabi-Sabi UX Concepts. Muted aesthetics and organic, dynamic data driven visuals can create meaningful experiences for users and help gain their loyalty, by:

- Choosing abstraction over granularity
- Choosing subtle over loud
- Choosing organic over structured
- Choosing incomplete over complete
- Choosing evolving data-backed visual elements

2.4 Wabi-Sabi Supports Transience and Uniqueness

The Scene: In the evening, Medkit gives Sam a birthday gift.

After a long and productive day, Sam is wrapping up his work. The clock reads 7:30 PM. His briefcase is open on his desk and he is looking at his tablet before packing it.

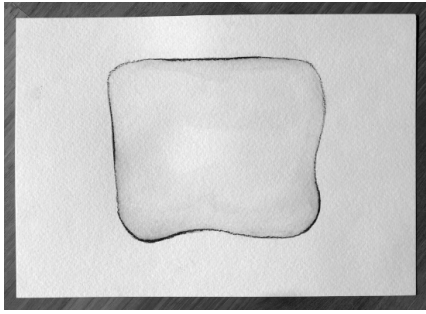


Figure 5: Wabi-Sabi embraces modest aesthetics

Medkit: This is a voluntary pulse check of your well-being. How do you feel?

Sam (tired): (sees 7 emoticons on the screen and taps two) Happy and Tired.

Medkit: You've had a long day! Would you like to take fewer appointments tomorrow?

Sam: Yes, just for tomorrow.

Medkit: You've earned it. While our product's algorithm wants to help you work efficiently, it also wants to help you stay happy and positive.

Sam: Okay.

Medkit: One more thing, here is a personalised discounted birthday package for you: 'Easy-vision', 'Voice input'.

Sam (surprised): Great, this sounds useful! Especially with my increasing long-sightedness and recent carpal tunnel syndrome.

Medkit: We understand people have different needs at different ages and situations in life. It's our job to evolve with you.

Sam (happy): Sounds great! But with all these changes, I hope you're not planning a complete re-vamp of your app. At this point, I'm set in my ways and have tailored it to suit my needs over the years.

Medkit: Old is gold. We never let go of what works well for you. No worries!

2.4.1 Derived Wabi-Sabi UX Concepts. A unique experience that caters to age and temporary contextual situations can deeply connect with the user's well-being, by:

- Using Artificial Intelligence (AI) for emotional well-being
- Providing age- and health-friendly features
- Supporting older releases
- Respecting user's preferences accumulated over years

2.5 Wabi-Sabi Co-creates with Nature

The Scene: In the Conference Room, Product X does the right thing.

Ten of Medkit's senior employees are gathered in a long conference room. Laptops are open and beverages are on the table. It's a big day, and everyone is eager to get started.

Senior Executive: We're a forward-looking company, and have always felt a sense of responsibility towards society and the world we live in.

Sustainability Officer: Our product is used by people throughout the world, and we have huge power to influence minds with our reach. We should use this sensibly, for the greater good.

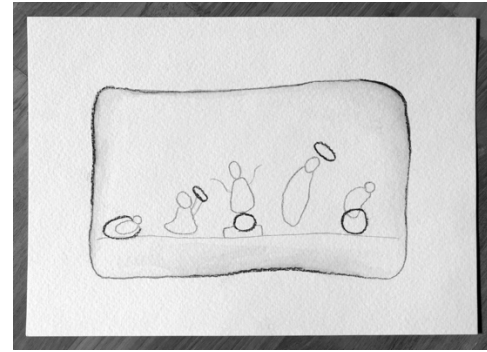


Figure 6: Wabi-Sabi supports transience and uniqueness

Product Owner: I agree, and this is exactly why we've invited your team to be part of our product discussions. We'd like to know if our ideas and processes have a negative impact on nature.

Designer: We all know our primary user group is doctors. Why don't we try the radical approach of having secondary personas for our product, like 'a tree', 'a bird' or 'a fish'? This will help us gain perspective on the problems we identify, and the ideas we churn out.

Developer: Interesting idea. It's true that technology can have a negative impact on our environment. We have to incorporate this awareness into the DNA of our product.

Designer: Can we include a 'sustainability indicator' against certain actions that impact the environment? This would help our users make informed decisions before they perform actions.

Product Owner: Yeah, we could also define a sustainability threshold for our product, and a maximum number of times it can be surpassed. If this is frequently exceeded, we can add it to our product backlog for investigation. If a user consistently opts for sustainable ways for doing business, we could offer them limited free trials for our product's new features.

Senior Executive: That's a win-win. While we subtly encourage our customers to opt for sustainable features, we can also promote new product features.

Sustainability Officer: I'm glad we're taking sustainability seriously. It's often sidelined nowadays, in the blind rush to innovate.

Product Owner: Yeah, innovation also has a high risk of errors and breaking smooth operations for our users, not to mention more effort and energy consumption for the makers. We always attempt to prioritise repairability, upgradability and renewability over radical innovation.

2.5.1 Derived Wabi-Sabi UX Concepts. Companies can promote sustainable attitudes, perceptions and behaviours with the inclusion of nature as a product factor, by:

- Looking at products through the lens of nature
- Including environmentally friendly processes, principles and ideas in their products
- Making the impact of user's actions on the environment transparent
- Allowing for slow coding
- Encouraging reuse, and eliminating re-inventing

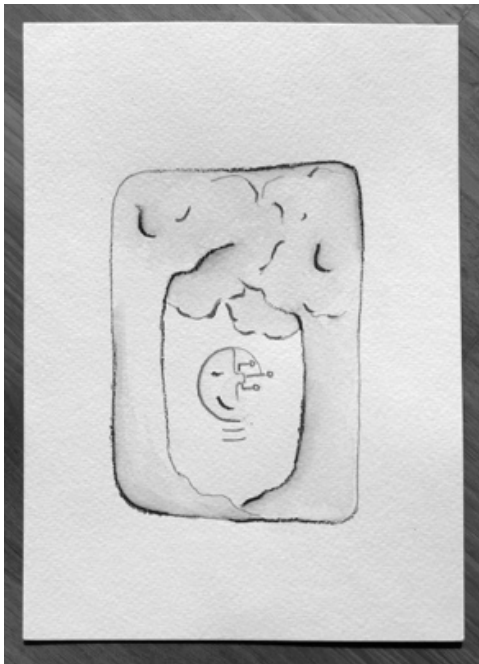


Figure 7: Wabi-Sabi co-creates with nature

3 FINDINGS

Prior research explored the intersections between Wabi-Sabi elements and the design of physical products, architecture, interior design and, on a generic level, UX design. Our derived Wabi-Sabi UX concepts show, in more granular ideas, their potential to influence digital user experiences, especially in business software.

The research method proved effective, and could be extended to investigate unconventional challenges that are not driven by user needs, but rather by intuition or notions of value. This approach allowed the translation of a vague possibility to concrete concepts, and clarified which aspects require further investigation in future research. It also helped assess a new idea's potential before monetary investments were made in traditional user research activities, such as recruiting users for participatory workshops.

Many of the speculations and derived UX concepts imply that there is a co-creation between a machine, software product, and human [15]. 'Learning from user habits' is one example in which the human allows the machine to learn from them. This requires respecting both ethical and data privacy guidelines, and involves detailed discussion about intellectual property rights, which are not discussed in this paper, and which require further research.

The feasibility and viability of these derived Wabi-Sabi UX concepts remain to be validated. At the time of writing, participatory workshops with designers tasked to prototype the concepts were planned, but had not yet been conducted. Further, detailed validations with end users, digital product providers and companies using the Wabi-Sabi inspired designs need to be conducted, and the results compared to the pre Wabi-Sabi versions of the user interface designs. Further research should also be directed at how to

determine the possible business impact of Wabi-Sabi inspired user interfaces.

4 CONCLUSION

Space. Space in an increasingly digital, noisy and complex world. A space to be human beckons. To breathe, to err, to be appreciated. To be born into technology, to be able to trust it and to grow old with it. To be imperfect, to be explorative. To stumble upon discoveries. To reflect. To experiment. To evolve. To be kind, to yourself and to the natural environment. To be part of systems and build products, with forward thinking businesses that are not ruthless in their economic growth, are not afraid of making mistakes, and don't turn a blind eye to their surroundings. To walk off the beaten path. To not lose track of experiences built on fundamental human needs.

To take small steps in the right direction. Steps that pave the way to a future where businesses, people and nature thrive in a thoughtful, forgiving and transient space – a perfectly imperfect space.

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